

help needed with tf.reduce_mean()

<https://github.com/tensorflow/quantum/issues/273>

ROW 5

help needed with tf.reduce_mean() #273

Closed



aswanthkrishna opened on Jun 22, 2020

I have a 2D tensor of circuits. I wish to find the expectation values of an operator for these circuits, and take the mean along an axis.

I have tried using `tf.reduce_mean()`.

it is giving me this error.

```
programs must be rank 1. Got rank 2. [Op:TfqSimulateExpectation]
```

```
#my model
result = tf.reduce_mean(
    tfq.layers.Expectation()(circuits,
                             symbol_names = symbols,
                             operators = Paulis,
                             initializer=tf.keras.initializers.RandomUniform(0, 2 * np.pi)),axis = 1)
```

am I doing something wrong?

Thanks in advance!



zaqqwerty on Jun 22, 2020 · edited by zaqqwerty

Edits Contributor ...

Hi, our circuit execution layers currently accept either a single Cirq circuit or a 1D list or tensor of circuits. To emulate higher dimensional circuit inputs, you can use the [tf.reshape](#) op on the output of our quantum layers.

Here is a code example to demonstrate the reshaping:

```
q = cirq.GridQubit(0, 0)
circuit_list_2d = [[cirq.Circuit(cirq.X(q)), cirq.Circuit(cirq.H(q))],
                   [cirq.Circuit(cirq.Y(q)), cirq.Circuit(cirq.S(q))]]

# Flatten to 1D
flat_circuit_list = []
for sublist in circuit_list_2d:
    flat_circuit_list += sublist
expectations_1d = tfq.layers.Expectation()(
    flat_circuit_list, symbol_names=[], operators=cirq.Z(q))
print(expectations_1d)

# Expand back to 2D
expectations_2d = tf.reshape(expectations_1d, [len(circuit_list_2d), len(circuit_list_2d[0])])
print(expectations_2d)

result = tf.reduce_mean(expectations_2d, axis=1)
print(result)
```



MichaelBroughton assigned [zaqqwerty](#) on Jun 22, 2020



aswanthkrishna on Jun 22, 2020

Author ...

Thank you [@zaqqwerty](#) . can i train such a model?

